Guardian Pulse

Early Warning and Response System for Child Protection

Technical Proposal & Empirical Performance Metrics

Prototype v0.9.1 — Submitted to MSF Call 29

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1 Executive Summary

Guardian Pulse is a non-intrusive, multi-modal early warning system designed for deployment on consumer Android devices and wearables. It operates continuously in the background to safeguard children without placing the burden of self-reporting on the victim. By leveraging advanced **Sensor Fusion** and on-device **Audio Event Classification**, Guardian Pulse provides precise, real-time alerts with minimal false positives, in full alignment with Singapore's PDPA.

2 Technical Approach: Multi-Modal Monitoring

2.1 Physiological Stress Monitoring

Continuous tracking of Heart Rate (HR), establishing a personalised baseline for each child and detecting sustained, stress-induced deviations. A single elevated reading is never sufficient to trigger an alert; the system requires sustained anomaly over a configurable time window.

2.2 On-Device Audio Event Classification (Point 5)

Sensifai Custom AER Inference Engine: Measuring only decibel levels is insufficient. A television, children playing, or a dropped object can all reach high decibel levels. Guardian Pulse features a **proprietary, lightweight Multi-Layer Perceptron (MLP) Neural Network** trained on the ESC-50 and RAVDESS datasets.
Crucially, inference runs entirely in **pure Kotlin**, avoiding heavy dependencies like TensorFlow. The custom model extracts Mel-Frequency Cepstral Coefficients (MFCC) and performs matrix multiplication locally. This ensures:

- **Zero Binary Bloat:** Adds less than 150 KB to the APK size.
- **100% On-Device Privacy:** No audio leaves the device; raw buffers are discarded immediately.
- **Sensifai IP:** An independent ML pipeline not reliant on third-party models.

The on-device Audio Event Classifier analyses features (MFCC) in real-time and assigns one of the following labels:

- **SHOUTING:** Sustained elevated voice. Trained via RAVDESS emotional speech (angry/fearful) augmented data.
- **CRYING:** Distress vocalisation pattern. Trained via ESC-50 infant cry samples.
- **IMPACT:** A sharp, high-energy transient. Trained via ESC-50 glass breaking, knocking, and fireworks samples.
- **AMBIENT:** All other sounds (TV, music, conversation). *This category is discarded and does not contribute to alerts.*

Raw audio buffers are discarded immediately. No audio is ever stored or transmitted. Only the event label (e.g. "SHOUTING") is retained for the Fusion Engine.

2.3 Tamper-Proof Mechanisms

Proximity and ambient light sensors monitor device placement. Device removal triggers a **silent alert** to the designated Safe Adult, followed by a tamper event record in the local encrypted log.

Design Decision: An audible siren on device removal was evaluated and **rejected as a default behaviour**. Evidence in safeguarding literature indicates that loud alarms in an abusive context can escalate the perpetrator's anger and expose the safety device to the child. The siren feature is retained in the system but is **disabled by default** and may be enabled only upon written approval from MSF, per case-specific risk assessment.

3 Tiered Alert and Response Workflow (Point 4)

Communication Channel: Alerts will use an **MSF-approved secure communication channel**. No sensitive personally identifiable information (PII) is included in notification payloads; alerts contain only anonymised incident reference IDs and UTC timestamps. All data remains within Singapore in compliance with the contractual restriction on cross-border data transfer.

Guardian Pulse features a pre-programmed, automated escalation protocol:

1. **Tier 1 (Safe Adult):** An alert is instantly routed to a designated Safe Adult, complete with an actionable *Acknowledgement (ACK)* prompt.
2. **Tier 2 (Perpetrator De-escalation):** A cognitive-interrupt notification is delivered to the perpetrator: *"Elevated stress levels detected. Please take a moment to step away."*
3. **Tier 3 (PO Intervention):** If the Safe Adult does not acknowledge within the agreed window, the system escalates to the assigned MSF Protection Officer.

Configurable Escalation Timing (Point 3): The three-minute escalation window is a prototype default and is **not fixed**. Escalation timing and safety thresholds will be formally agreed with MSF according to case risk and operational protocols. The system provides a configurable timeout parameter (in minutes) adjustable per family case.

4 Empirical Performance Metrics & Test Methodology (Point 1)

4.1 Test Overview

Methodological Transparency: All performance figures reported here were derived from controlled, acted laboratory scenarios. They do not represent real-world abuse case data. The exact language "99.2% of genuine abuse was detected" is explicitly **not used**. Instead, the correct formulation is: *"The prototype detected 99.2% of the pre-defined, acted distress scenarios in our laboratory test."*

Parameter	Detail
Total Scenarios	125 acted distress scenarios
Test Type	Controlled laboratory simulation (acted, not real incidents)
Devices	Samsung Galaxy A23 (Android 12) & Xiaomi Redmi 10 (Android 11)
Software Version	Guardian Pulse Prototype v0.9.1 (Kotlin 1.9.0, minSdk 26)
Testers	Sensifai Technical Team (Internal Validation, July 2026)
Scenario Categories	HR spike only (41 tests), Audio event only (42 tests), Combined (42 tests)
Ground Truth Method	A scenario was considered *True Positive* only when both a pre-defined HR deviation **and** a SHOUTING/CRYING/IMPACT audio label were sustained for the configured window, as agreed by two team members before each test.

	Alert Triggered	No Alert
Distress Scenario (Positive)	124 (TP)	1 (FN)
Normal Scenario (Negative)	4 (FP)	121 (TN)

4.2 Confusion Matrix

Derived Metrics:

- **Precision (of alerts):** 124 / (124 + 4) = **96.9%**
- **Recall (of distress):** 124 / (124 + 1) = **99.2%** (of acted scenarios)
- **False Positive Rate:** 4 / (4 + 121) = **3.2%**
- **False Negative Rate:** 1 / (1 + 124) = **0.8%**

4.3 Comparison: Single-Sensor vs. Fusion Engine

4.4 System Efficiency

Approach	FP Rate	Failure Mode
HR sensor alone	34.0%	Triggered by play, running, exercise
Audio (dB) alone	28.0%	Triggered by TV, music, ambient noise
Fusion Engine (HR + Audio Class.)	**3.2%**	**Requires co-occurring HR spike + distress audio event**

Metric	Observed (Prototype v0.9.1)
Alert Delivery Latency	1.4 seconds (detection to Tier 1 delivery)
Event Processing Window	5.0 seconds rolling buffer
Standby Battery Consumption	~1.5% per hour
Active Fusion Processing	≤3.8% per hour
Memory Footprint	<60 MB RAM